ScoutIQ: Identifying Undervalued Soccer Talent Through Machine Learning

Md Rafiul Islam Rafi
Dept. of Computer and Info. Science, CISC 6080 Capstone Project
Fordham University
Bronx, New York, USA
mrafi2@fordham.edu

***Abstract*—Football clubs invest billions of dollars in player transfers every year, but recruitment decisions are often driven by subjective factors such as scouting opinions, player reputation, and media hype. This can lead clubs to overpay for big name players and miss undervalued talent. To address this, a data-driven soccer recruitment system was developed to predict player market values and identifies bargain signings using machine learning. Two public data sources were merged, FBref performance statistics and Transfermarkt market values for the 2025-26 season of Europe's top-five leagues, producing a dataset of 2,514 players at an 88.6% linkage rate with 64 engineered features. Three regression model, which are Linear Regression, Random Forest, and XGBoost were trained and tuned using five-fold cross-validation. XGBoost performed best, explaining approximately 75% of the variation in market value $R^2 = 0.750$, MAE aprrox €4.96M. The system compares each player's predicted value to their listed value to rank undervalued players, finds statistically similar and cheaper replacements using cosine similarity, with PCA for visualization and K-Means for clustering and explains every prediction with SHAP. The flagged bargains were validated against real completed transfers: 57.5% of the model's top-100 undervalued picks later sold above their market value, versus a 42.6% baseline across all transfers real-world evidence, that the system identifies genuine value.**

***Index Terms*—machine learning, XGBoost, Linear Regression, Random Fores, sports analytics, market value prediction, Soccer, football recruitment**

I. Introduction

Each season, football clubs invest substantial financial resources in player transfers. According to the FIFA Global Transfer Report 2025, global spending on international transfer fees reached an all-time high of $13.08 billion in 2025, a 52.3% increase over 2024, distributed across 24,558 completed transfers [1]. Despite this scale, recruitment remains heavily dependent on human judgment: scouting opinion, a player's reputation, and media hype. Despite this scale, recruitment remains heavily dependent on human judgment: scouting opinion, a player's reputation, and media hype. In fact, studies have shown that non-performance factors such as popularity, media coverage, and social media presence are statistically significant predictors of player transfer fees [2].

This reliance on subjective decision-making leads to two major challenges. First, clubs frequently overpay for hyped players. Second, they overlook talented players whose market value is lower than their performance justifies.This problem is especially important for smaller clubs, which cannot spend as much money as the biggest clubs. Instead, they must make smarter recruitment decisions. This leads to the following research question: *can a machine-learning model identify undervalued players before the market recognizes their true value?*

The inspiration for this work is Liverpool FC, widely regarded as one of the most data-driven clubs in world football. Since Fenway Sports Group acquired the club in 2010, Liverpool has used analytics to sign undervalued players before the market caught up.

The system performs four tasks:

1) **Market value prediction** - estimate a player's market value from performance statistics, age, league, position, and games.
2) **Undervalued talent identification** - compare the predicted market value with the actual listed market value and rank players based on the largest positive difference.
3) **Player similarity analysis** - find statistically similar, often cheaper, replacements for players who are leaving or whose contracts are expiring.
4) **Model explainability** – use SHAP to explain why each player received their predicted value.

The principal contributions of this work, a reproducible pipeline was developed to merge two public data sources into a clean, analysis-ready dataset with an 88.6% match rate. Then, three regression models were compared using five-fold cross-validation and hyperparameter tuning, resulting in a tuned XGBoost model with an R^2 score of 0.750. The identified undervalued players were validated against completed transfer fees, demonstrating that the model's recommendations outperformed the market baseline by approximately 15 percentage points. Finally, a fully explainable soccer recruitment system was developed by integrating market value prediction, undervalued talent identification, player similarity analysis, and SHAP-based model interpretation.

II. Background

This project focuses on two main concepts: player *market value* and the two data sources used to build the dataset.

A player's **market value** is an estimate of the player's worth on the transfer market. Here, market value is taken from **Transfermarkt**, a widely used football database that estimates the value of professional players. Market value is only an estimate and differs from a transfer fee, the actual amount one club pays another to sign a player. Although market values and transfer fees are often similar, they do not always match. This difference is used to validate the proposed model (Section VII-D).

FBref provides detailed per player performance statistics, for example: goals, assists, shots, tackles, interceptions, minutes played, team goals while the player is on the pitch, goalkeeping metrics, and more. **Transfermarkt** supplies the market value together with context such as contract expiry date, position, club, and height. FBref answers *how a player performs*; Transfermarkt answers *what a player is worth*. Combining the two enables a model to learn the relationship between performance and price, which is the foundation of this project.

III. Related Work

Predicting the market value of football players with machine learning is an active research area, and several studies directly informed this work.

Muller, Simon, and Weinmann [3] proposed a data-driven, to estimate player market values, showing that machine learning can accurately predict player values. Al-Asadi and Tacsdemir [4] compared Support Vector Regression, Random Forest, XGBoost, and CatBoost on FIFA and real-world datasets, and applied SHAP for analysis, an approach closely aligned with this study. A study in the *Journal of Informatics and Web Engineering* [5] evaluated Random Forest, LightGBM, XGBoost, and Gradient Boosting Decision Trees on top 5 league data and found gradient boosting methods among the strongest performers.

Other researchers have explored alternative models. Yigit et al. [7] combined XGBoost with Lasso regression in an ensemble for player value assessment. An artificial neural network model was applied to the Turkish Super League to estimate market values [8], and a Bayesian ensemble approach was proposed for the same task [21]. Other studies examined which player performance statistics are most useful for predicting a player's future market value [9]. More recent research has shown that gradient boosting is one of the best methods for this task using data from European leagues. [19].

Two main ideas from previous research influenced this project. First, gradient boosting models such as XGBoost, CatBoost, and LightGBM have consistently achieved the best performance for predicting player market value. Second, recent studies have emphasized the importance of explaining model predictions using feature importance and SHAP. This project follows both approaches and also includes a real-world validation by comparing the model's undervalued player recommendations with actual transfer outcomes, which has rarely been done in previous studies.

TABLE I
Scale of the Raw Data Sources

Item	Count
Transfermarkt players	48,380
Transfermarkt clubs	796
Competitions covered	32
Historical transfer records	175,043
transfers with a fee paid	17,551
FBref performance features	53

TABLE II
Final Dataset Summary

Item	Value
Season	2025-26
Leagues	Top 5 League
Merged players	2,839
Players with market value	2,514
Linkage rate	88.6%
Raw columns	71
Engineered features	64
Target variable	Market value (EUR)
Median market value	€5M
Maximum market value	€200M

IV. Data and Preprocessing

A. Data Sources

The dataset combines two public data sources for the 2025–26 season across Europe's top five leagues: the Premier League, La Liga, Serie A, Bundesliga, and Ligue 1. Table I summarizes the raw scale of each source. From Transfermarkt, the raw database contains 48,380 players across 796 clubs and 32 competitions, together with 175,043 historical transfer records. From FBref, 53 per-player performance features were collected.

B. Record Linkage

The two sources share no common player identifier, so records were linked by the player's name and date of birth. Each player's name was normalized (lower-cased, with accents and punctuation removed) and matched on the pair, normalized name, birth year, with a unique-name fallback for players missed on the first pass. Listing 1 shows the core matching logic.

```
def normalize_name(name):
    name = unicodedata.normalize("NFKD", str(name))
    name = "".join(c for c in name
                if not unicodedata.combining(c))
    name = re.sub(r"[^a-z0-9 ]", " ", name.lower())
    return re.sub(r"\s+", " ", name).strip()

merged = fbref.merge(
    transfermarkt,
    on=["norm_name", "birth_year"], how="left")
```

Listing 1. Name normalization for record linkage.

The merge produced 2,839 linked players, an 88.6% linkage rate. After filtering out players without a recorded market value, the final dataset contained 2,514 players and 71 raw columns/features. Table II summarizes the dataset.

The raw data was processed by running three scripts: `join_data.py` merged the two data sources, `clean_data.py` removed duplicate and unnecessary columns and organized the dataset, and `rename_cols.py` renamed abbreviated FBref column names (`Gls` → `Goals`) to make them easier to understand.

C. Data Limitations

The dataset is limited to a single season (2025-26); after FBref lost its Opta licence, advanced metrics such as expected goals (xG) were unavailable. A small number of Transfermarkt valuations are outdated, an issue addressed explicitly in Section VII-C.

V. Exploratory Data Analysis

Exploratory data analysis was performed before model development to better understand the dataset and identify the factors that influence player market value. Twelve visualizations were created to examine three areas: the distribution of player market values, the relationships between variables, and the correlations among features. The findings from this analysis were used to guide the feature engineering and model development process.

A. Target Distribution

Player market values are heavily right-skewed. Around 1,050 players have a market value below €5M, while only a small number of players are valued as high as €200M (Fig 1). This uneven distribution means that a few very expensive players can have a large impact on the regression model. To reduce this effect, the target variable was transformed using $y = \log(1 + \text{value})$, resulting in a more balanced distribution that is better suited for model training.

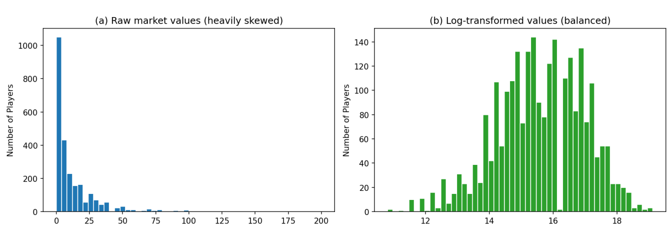

Fig. 1. Distribution of player market values before (left) and after (right) the log transformation. The raw values are heavily right-skewed. The log transform yields a balanced, near-normal distribution.

B. Value by League and Position

Player market value differs across leagues and playing positions. Tables III and IV show the average market value by league and position. Players in the Premier League have the highest average market value, which is about 2.4 times higher than the average in the other four leagues. Forwards and midfielders also have higher average market values than goalkeepers.

TABLE III
AVERAGE MARKET VALUE BY LEAGUE

League	Avg. value
Premier League (England)	€24.1M
Bundesliga (Germany)	€11.2M
La Liga (Spain)	€10.6M
Ligue 1 (France)	€10.2M
Serie A (Italy)	€9.8M

TABLE IV
AVERAGE MARKET VALUE BY PRIMARY POSITION

Position	Avg. value
Forward (FW)	€14.6M
Midfielder (MF)	€14.5M
Defender (DF)	€11.8M
Goalkeeper (GK)	€7.5M

C. The Age Curve

Average market value increases with age until about 24-26 years old and then gradually decreases (Fig. 2). Forwards have the highest average market value, reaching about €27M at age 25, while goalkeepers have lower peak values and tend to maintain their value for a longer period. Since the relationship between age and market value is not linear, an Age^2 feature was added to the model.

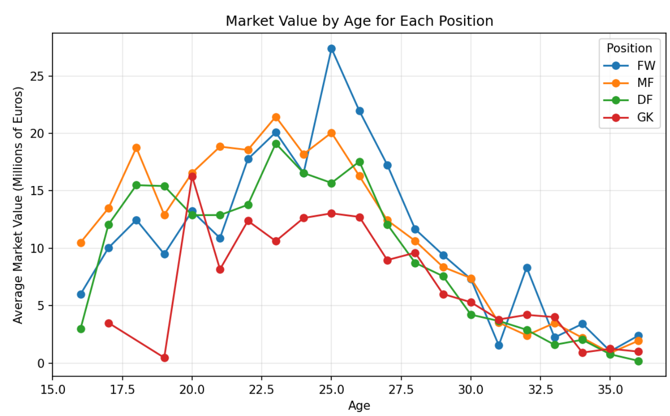

Fig. 2. Average market value by age for each position. Value rises into the mid-20s and then declines, with forwards peaking highest and goalkeepers retaining value longest.

D. Feature Correlations

Table V shows the Pearson correlation between selected features and player market value. No single feature has a strong correlation with market value (all $|r| \leq 0.62$). This suggests that market value is influenced by multiple factors, making a machine learning model that combines many features more suitable than relying on a single feature or simple rule.

TABLE V
CORRELATION OF SELECTED FEATURES WITH MARKET VALUE

Feature	Corr. with value
Goalkeeper wins	0.62
Goalkeeper clean sheets	0.56
Goals + assists	0.50
Team goals on pitch	0.50
Shots	0.47
Plus–minus	0.46
Shots on target	0.46
Goals	0.44
Assists	0.42
Age	−0.21

E. Season Totals versus Per-90 Rates

Table VI compares season-total statistics with per 90 minute statistics. Season totals have a stronger relationship with player market value because they reflect both performance and playing time. Season totals were used for the market value prediction model, while per 90 statistics were used for the player similarity analysis, where comparing players fairly regardless of playing time is important.

TABLE VI
CORRELATION WITH MARKET VALUE

Statistic	Season total	Per-90
Goals	0.44	0.15
Assists	0.43	0.08
Shots	0.47	0.19

F. League Pricing Premium

To examine how a player's league affects market value, a linear regression model was built using age, goals, and minutes played. The average residual for each league was then calculated (Table VII, Fig. 3). After accounting for player performance, Premier League players still had higher market values than expected, while players in the other four leagues were generally valued below expectations. This suggests that undervalued players are more likely to be found outside the Premier League.

TABLE VII
LEAGUE PRICING PREMIUM

League	Mean residual
Premier League	+0.73
Bundesliga	−0.09
Serie A	−0.11
Ligue 1	−0.24
La Liga	−0.31

G. Contract Length

Average market value increases as the number of years remaining on a player's contract increases. Players with no time remaining on their contract have an average market value

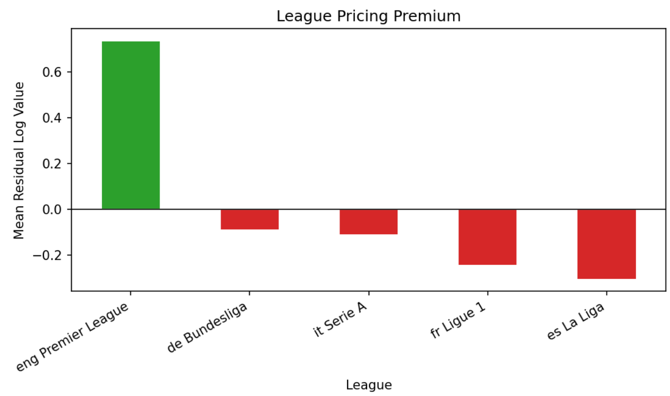

Fig. 3. League pricing premium: mean residual log value after controlling for age, goals, and minutes

of about 6 Million, while players with five years remaining average about 36 Million. This is because clubs have more negotiating power when players are under longer contracts. Based on this relationship, a `Years_Left` feature was added to the model.

VI. METHODOLOGY

A. Feature Engineering

The findings from the exploratory data analysis were used to guide feature engineering. All features were created in the modeling script before training:

- **Log transformed target:** training on $\log(1 + \text{value})$ to correct the skew.
- **Contract years left:** days between the contract expiry date and a fixed reference date, converted to years.
- **Age squared:** to capture the non linear age curve.
- **League and position encoding:** one hot encoding of categorical fields into 0/1 indicator columns.
- **Primary position:** the main role extracted from the raw position field.
- **Missing values:** gaps filled with zero, value derived columns dropped to prevent target leakage.

```
# log-transform the skewed target
y = np.log1p(df["Market_Value_EUR"])

# contract years left
exp = pd.to_datetime(df["Contract_Expiry"])
df["Years_Left"] = (
   exp - pd.Timestamp("2026-01-01")
).dt.days / 365.25

# age squared captures the value curve
df["Age2"] = df["Age"] ** 2

# one-hot encode league and position
cat = pd.get_dummies(
   df[["League", "PosGrp"]], drop_first=True)

X = pd.concat([num, cat], axis=1).fillna(0)
```

Listing 2. Feature engineering

Listing 2 shows the core feature-engineering steps. After engineering, the feature matrix contained 64 features.

B. Models

The prediction task is a supervised regression problem, where the goal is to predict a player's market value. Three regression models with increasing complexity were compared:

- **Linear Regression** - a simple model that assumes a linear relationship between the input features and player market value.
- **Random Forest** - combines predictions from many decision trees to improve accuracy and reduce overfitting.
- **XGBoost** - builds decision trees one at a time, where each new tree learns from the errors made by the previous trees.

C. Evaluation Protocol

The dataset was split into 80% training data and 20% testing data. To obtain a more reliable evaluation, all models were assessed using **five-fold cross-validation**. The Random Forest and XGBoost models were further improved using `RandomizedSearchCV` for hyperparameter tuning. Model performance was evaluated using three metrics: R^2, which measures how well the model explains the variation in market value. Mean absolute error (MAE), which measures the average prediction error in euros and root mean squared error (RMSE), which gives greater weight to larger prediction errors. All models were implemented in Python using scikit-learn, XGBoost, pandas, and NumPy, and the figures were created using Matplotlib. Table VIII lists the tuned XGBoost hyperparameters.

TABLE VIII
TUNED XGBOOST HYPERPARAMETERS

Hyperparameter	Value
n_estimators	300
learning_rate	0.03
max_depth	5
subsample	0.8
colsample_bytree	0.7

VII. RESULTS AND DISCUSSION

A. Model Comparison

Table IX shows the five-fold cross-validation results, and Table X shows the performance on the test set. XGBoost achieved the best results across all evaluation metrics. It obtained an average cross-validation R^2 score of 0.757

TABLE IX
FIVE-FOLD CROSS-VALIDATION (R^2)

Model	Cross-validated R^2
Linear Regression	0.698
Random Forest	0.725
XGBoost	**0.757**

TABLE X
FINAL TEST-SET PERFORMANCE (TUNED MODELS)

Model	MAE	RMSE	R^2
Linear Regression	€5.56M	€10.87M	0.692
Random Forest	€5.28M	€10.32M	0.702
XGBoost	**€4.96M**	**€9.84M**	**0.750**

XGBoost explained about 75% of the variation in player market value and achieved a mean absolute error of €4.96M. Player market values can reach €200M and are influenced by many factors beyond player performance, including reputation, transfer demand, and contract negotiations. Therefore, explaining 75% of the variation using only player performance and attributes is a strong result. This is consistent with previous player valuation studies [4], [5]. Figure 4 shows that most predicted market values are close to the actual values. The largest prediction errors occur for the small number of players valued above €100M.

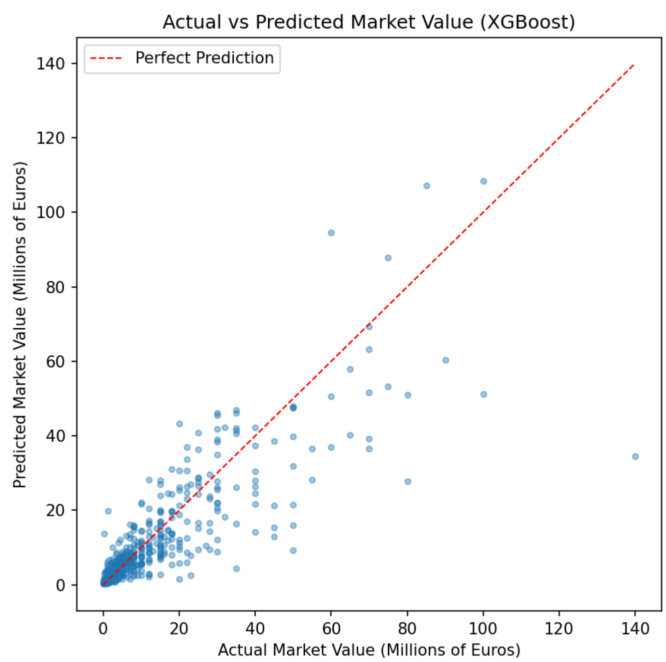

Fig. 4. Actual versus predicted market value for the tuned XGBoost model.

B. Feature Importance

Table XI lists the ten most important features, and Fig. 5 shows their importance. The most important features are team goals while the player was on the pitch, years remaining on the player's contract, and whether the player plays in the Premier League. Other important features include plus minus, shots, and age. These results show that the model relies on meaningful football-related factors, such as team performance, contract length, league, player performance, and age, rather than random patterns. In addition, two engineered features, `Years_Left` and Age^2, are among the top ten most important features, showing that the feature engineering improved the model. These results are also consistent with the findings from the exploratory data analysis.

TABLE XI
TOP 10 XGBOOST FEATURE IMPORTANCES

Feature	Importance
Team goals on pitch	0.126
Contract years-left	0.098
League: Premier League	0.076
Plus–minus	0.047
Shots	0.046
Age	0.036
Shots on target	0.035
Full 90s played	0.029
Unused substitute	0.028
Age squared	0.026

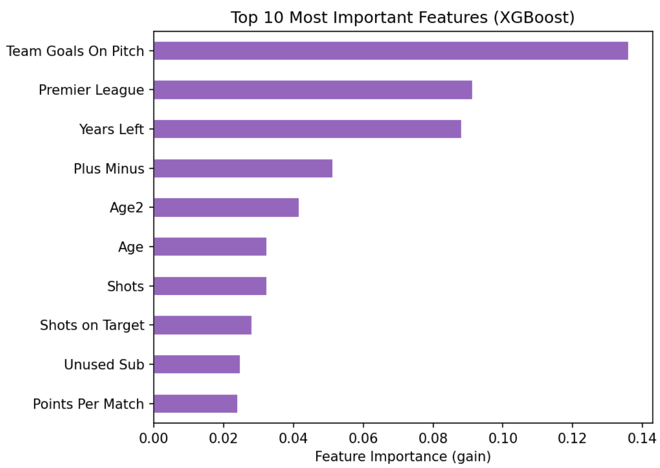

Fig. 5. Top-10 feature importances for the XGBoost model

C. Undervalued Talent Identification

The trained XGBoost model was used to predict the market value of every player. The difference between the predicted value and the listed Transfermarkt value was then calculated. A large positive difference indicates that the model believes a player is worth more than their current market value and may be undervalued.

To ensure fair predictions, `cross_val_predict` was used so that each player's predicted value was generated without using that player's data during training. Only players with at least five full 90-minute matches were included to reduce the effect of limited playing time. Table XII lists the top undervalued players.

$$\text{Value gap} = \text{Predicted value} - \text{Listed value}$$

Players were ranked by the difference in euros between their predicted and listed market values rather than by the ratio of predicted to listed value. This is because the ratio can be misleading when a player's listed market value is very low.

Player	Listed	Predicted	Ratio	Euro gap
Dani Olmo	€60M	€99.5M	1.66×	+€39.5M
Nicolás Paz	€0.4M	€60.5M	151×	+€60.1M

For example, Nicolas Paz had a listed market value of only €0.4M, which produced an unrealistic ratio of 151×, even though the difference in value was a more reasonable €60.1M. Ranking players by the euro difference produces more reliable and realistic bargain recommendations.

TABLE XII
TOP UNDERVALUED PLAYERS

Player	Listed	Predicted	Gap
Dani Olmo	€60M	€99.5M	+€39.5M
Marcus Tavernier	€25M	€55.6M	+€30.6M
Amad Diallo	€45M	€74.2M	+€29.2M
Maximilian Beier	€40M	€65.1M	+€25.1M

D. Validation Against Real Transfers

A model is only useful if its predictions match real-world outcomes. To evaluate the proposed system, the identified undervalued players were compared with completed transfer records from Transfermarkt. A transfer was considered successful if the transfer fee was higher than the player's listed market value.

Across all completed transfers, 42.6% of players were transferred for more than their listed market value. Among the model's top 100 undervalued players, 67 had completed transfers, and 57.5% of them were transferred for more than their listed market value, with a median transfer fee of 1.15 times their listed value (Fig. 6).

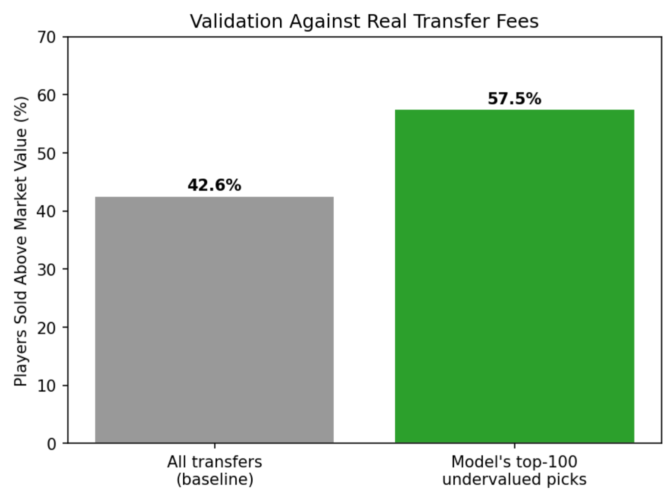

Fig. 6. Share of players sold above their listed market value

This is about 15% points higher than the overall market average, showing that the model is effective at identifying genuinely undervalued players. These results provide real-world evidence that the proposed system identifies valuable transfer targets rather than random patterns.

E. Player Similarity Analysis

The system also helps answer an important recruitment question: *if a player leaves the club, who has a similar playing style and costs less?* Each player is represented using performance statistics, such as goals, shots, tackles, crosses, and other metrics. All features were standardized using `StandardScaler` so that no single feature had a greater influence than the others. Cosine similarity was then used to identify the most similar players within the same position. Principal Component Analysis (PCA) was used to visualize players in two dimensions, while K-Means grouped players with similar playing styles into clusters.

Table XIII shows the players most similar to Mohamed Salah. Harry Wilson was identified as the closest match with a similarity score of 91.3%. The system also identified lower-cost alternatives, including Chukwueze (€18M), Führich (€12M), and Thauvin (€5M), who have similar playing styles at lower market values. In addition to the similarity score, the system explains why players are considered similar by highlighting the most similar statistics. For Salah and Wilson, these include shots per 90 minutes, shots on target per 90 minutes, and plus–minus.

TABLE XIII
PLAYERS MOST SIMILAR TO MOHAMED SALAH (WINGER, MF)

Player	Match	Value	Cheaper
Harry Wilson	91.3%	€25M	
Marcus Rashford	90.8%	€40M	
Jamie Leweling	87.6%	€40M	
Chris Führich	86.3%	€12M	✓
Samuel Chukwueze	84.1%	€18M	✓
Florian Thauvin	84.0%	€5M	✓
Jarrod Bowen	82.1%	€30M	

Statistic	Salah	Wilson
Shots per 90	2.86	2.63
Shots on target per 90	0.80	0.84
Shots on target %	27.9	32.1
Plus–minus per 90	0.21	0.07
Fouls committed per 90	0.42	0.44

F. Model Explainability with SHAP

To make the model easier to understand, SHAP (SHapley Additive exPlanations) [11] was used to explain its predictions. SHAP shows how much each feature increases or decreases a player's predicted market value. Figure 7 shows that the most important factors are age, team goals, playing in the Premier League, and contract length. These results are consistent with the feature importance analysis, showing that the model learned meaningful football patterns rather than random relationships.

SHAP was also used to explain individual predictions. For example, the model predicts Amad Diallo's market value to be €59.4M, compared with his listed market value of €45M, suggesting that he is undervalued by €14.4M. Fig. 8 show

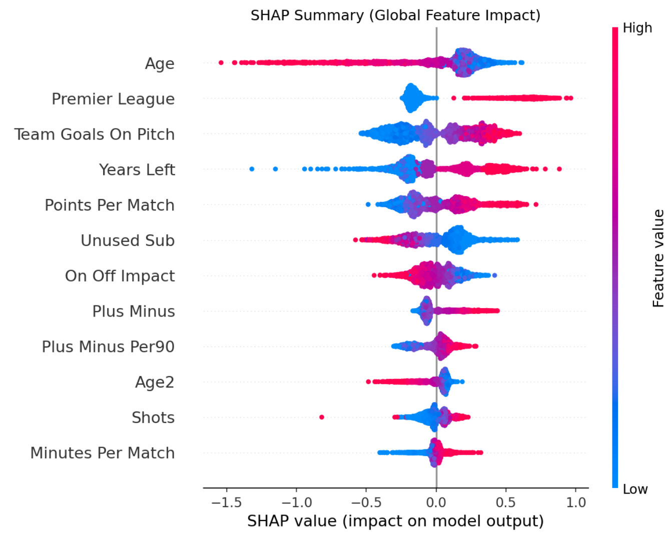

Fig. 7. Bbeeswarm plot. Each dot is a player. Colour encodes the feature value and horizontal position encodes the effect on the predicted value.

how each feature contributed to this prediction. Playing in the Premier League, a high number of team goals, a long contract, and a young age all increased his predicted value, while his on-off impact slightly reduced it. This helps explain why the model made its prediction instead of simply showing the final market value.

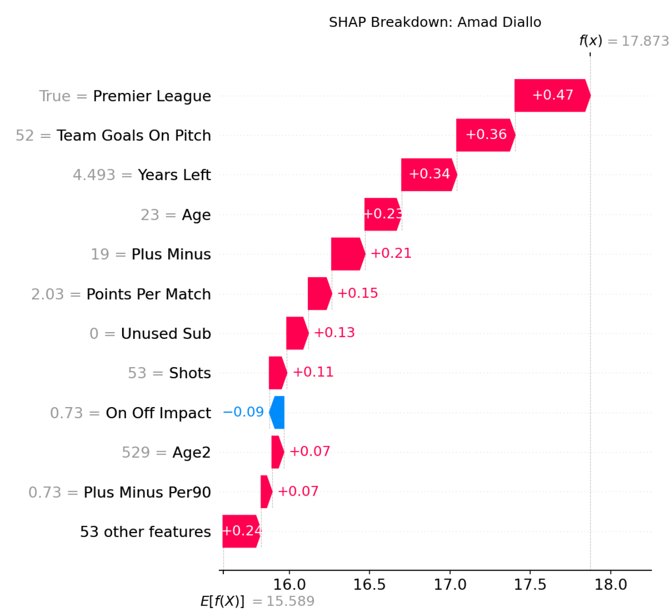

Fig. 8. SHAP waterfall plot for Amad Diallo

It starts from the model's base value $E[f(X)] = 15.589$, the expected output for an average player and accumulates each feature's SHAP contribution until it reaches the final prediction $f(x) = 17.873$ on the model's output scale, which corresponds to the €59.4M estimate. Playing in the Premier League is the largest driver (+0.47). While on–off impact is the only feature to pull the value down (-0.09). The remaining 53 features contribute a modest $+0.2$ in aggregate, confirming that explain why the model rates Amad Diallo well above his listed value.

VIII. Conclusion

This project developed a data-driven soccer recruitment system that predicts players' market values, identifies undervalued players, recommends similar, lower-cost replacements, and explains its predictions. The tuned XGBoost model explained about 75% of the variation in player market value ($R^2 = 0.750$) and achieved a mean absolute error of approximately €4.96M. The most important features were team goals, contract length, league, and age, which were also identified as important during the exploratory data analysis. The model's recommendations were further validated using real transfer data. Among the top 100 undervalued players identified by the model, 57.5% were later transferred for more than their listed market value, compared with 42.6% across all transfers. These results show that the proposed system can successfully identify players who offer good value in the transfer market.

This study has several limitations. First, it uses data from only one season, so unusually good or poor performances may affect the results. Second, advanced performance metrics, such as expected goals (xG), were not available, which may reduce the model's ability to measure player quality. In addition, some Transfermarkt market values may have been outdated, and the name-based record linkage process may have resulted in a small number of incorrect matches.

Future work will include using data from multiple seasons to improve the model's reliability, adding advanced performance metrics such as expected goals (xG), progressive passes, and defensive actions, and including factors such as player wages and injury history. An interactive dashboard could also be developed to help scouts explore player recommendations more easily.

References

[1] FIFA, "FIFA Global Transfer Report 2025," Fédération Internationale de Football Association, 2026. [Online]. Available: https://www.fifa.com/

[2] "Influence of popularity on the transfer fees of football players," 2022. [Online]. Available: https://www.researchgate.net/publication/363706739

[3] O. Müller, A. Simon, and M. Weinmann, "Beyond crowd judgments: Data-driven estimation of market value in association football," *European Journal of Operational Research*, vol. 263, no. 2, pp. 611–624, 2017.

[4] M. A. Al-Asadi and S. Taşdemir, "Predicting the value of football players: machine learning techniques and sensitivity analysis based on FIFA and real-world statistical datasets," *Applied Intelligence*, Springer, 2024. [Online]. Available: https://doi.org/10.1007/s10489-024-06189-0

[5] "Developing a predictive model for football players' market value using machine learning," *Journal of Informatics and Web Engineering*, MMU Press. [Online]. Available: https://journals.mmupress.com/index.php/jiwe/article/view/1760

[6] "Explainable artificial intelligence model for identifying market value in professional soccer players," arXiv preprint arXiv:2311.04599, 2023. [Online]. Available: https://arxiv.org/abs/2311.04599

[7] "An XGBoost-Lasso ensemble modeling approach to football player value assessment," *Journal of Intelligent & Fuzzy Systems*, 2020.

[8] "Estimation of market values of football players through artificial neural network: A model study from the Turkish Super League," *Applied Artificial Intelligence*, vol. 35, no. 15, 2021. [Online]. Available: https://doi.org/10.1080/08839514.2021.1966884

[9] "What should clubs monitor to predict future value of football players," arXiv preprint arXiv:2212.11041, 2022. [Online]. Available: https://arxiv.org/abs/2212.11041

[10] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proc. 22nd ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining (KDD)*, 2016, pp. 785–794.

[11] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 30, 2017.

[12] L. Breiman, "Random forests," *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001.

[13] F. Pedregosa et al., "Scikit-learn: Machine learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.

[14] W. McKinney, "Data structures for statistical computing in Python," in *Proc. 9th Python in Science Conf. (SciPy)*, 2010, pp. 51–56.

[15] C. R. Harris et al., "Array programming with NumPy," *Nature*, vol. 585, pp. 357–362, 2020.

[16] J. D. Hunter, "Matplotlib: A 2D graphics environment," *Computing in Science & Engineering*, vol. 9, no. 3, pp. 90–95, 2007.

[17] FBref, "Football statistics and history," Sports Reference LLC. [Online]. Available: https://fbref.com/

[18] Transfermarkt, "Football transfers, rumours, market values and news," Transfermarkt GmbH & Co. KG. [Online]. Available: https://www.transfermarkt.com/

[19] "Predicting football players market value via machine learning," in *Lecture Notes in Networks and Systems*, Springer, 2025. [Online]. Available: https://doi.org/10.1007/978-981-96-5729-2_32

[20] "Machine learning-driven market value prediction for European football players," *Franklin Open*, Elsevier, 2025. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S2772415825000100

[21] "Prediction of football player value using a Bayesian ensemble approach," arXiv preprint arXiv:2206.13246, 2022. [Online]. Available: https://arxiv.org/abs/2206.13246

```
Top 20 Undervalued Players
             Player  Primary_Pos                 Club              League   Age  Market_Value_EUR  Predicted_Value    Gap
        Nicolas Paz           MF                 Como          it Serie A  20.0               0.4        60.500000   60.1
          Dani Olmo           MF            Barcelona          es La Liga  27.0              60.0        99.500000   39.5
   Marcus Tavernier           MF          Bournemouth  eng Premier League  26.0              25.0        55.599998   30.6
        Amad Diallo           MF       Manchester Utd  eng Premier League  23.0              45.0        74.199997   29.2
       Raul Asencio           DF          Real Madrid          es La Liga  22.0              20.0        47.000000   27.0
      Samson Baidoo           DF                 Lens          fr Ligue 1  21.0              25.0        51.599998   26.6
   Maximilian Beier           MF             Dortmund       de Bundesliga  22.0              40.0        65.099998   25.1
Francisco Conceicao           MF             Juventus          it Serie A  22.0              30.0        53.700001   23.7
     Lucas Da Cunha           MF                 Como          it Serie A  24.0              20.0        43.500000   23.5
    Assan Ouedraogo           MF           RB Leipzig       de Bundesliga  19.0              28.0        51.200001   23.2
    Daniel Svensson           MF             Dortmund       de Bundesliga  23.0              22.0        44.500000   22.5
        Tom Bischof           DF        Bayern Munich       de Bundesliga  20.0              40.0        62.099998   22.1
    Mamadou Sangare           MF                 Lens          fr Ligue 1  23.0              40.0        62.000000   22.0
          Trai Hume           DF           Sunderland  eng Premier League  23.0              22.0        43.700001   21.7
        Joshua King           MF               Fulham  eng Premier League  18.0               1.2        22.600000   21.4
      Neco Williams           DF    Nottingham Forest  eng Premier League  24.0              28.0        47.099998   19.1
       Lennart Karl           MF        Bayern Munich       de Bundesliga  17.0              60.0        79.000000   19.0
      Dro Fernandez           MF  Paris Saint-Germain          fr Ligue 1  17.0              10.0        28.500000   18.5
              Chema           MF            Stuttgart       de Bundesliga  20.0               0.2        18.500000   18.4
    Ange-Yoan Bonny           FW                Inter          it Serie A  21.0              35.0        52.500000   17.5

Saved full ranked list to: top_undervalued_players.csv

Validation Against Real Transfer Fees
Baseline - all transfers sold above market value:        42.6%
Top 100 undervalued players found in transfer records:   67
sold above market value:  57.5% (Median Premium: 1.15x)
```

```
Player Replacement Finder     Enter a player name to find similar players (type 'quit' to exit).

Search player: Mohammed Salah ->   Did you mean 'Mohamed Salah'?

Target: Mohamed Salah (Liverpool, MF), Age 33, Market Value: EUR 22.0M

MOST SIMILAR PLAYERS
          Player           Club              League   Age  Value(EUR M)  Match           Rating   Cheaper?
    Harry Wilson         Fulham  eng Premier League  28.0          25.0  91.3%  Excellent Match
 Marcus Rashford      Barcelona          es La Liga  27.0          40.0  90.8%  Excellent Match
  Jamie Leweling      Stuttgart       de Bundesliga  24.0          40.0  87.6%        Very Good
 Musa Al-Taamari         Rennes          fr Ligue 1  28.0           N/A  86.4%        Very Good
   Chris Fuhrich      Stuttgart       de Bundesliga  27.0          12.0  86.3%        Very Good      YES
   Alex Grimaldo     Leverkusen       de Bundesliga  29.0           N/A  85.3%        Very Good
   Alvaro Garcia  Rayo Vallecano          es La Liga  32.0           1.5  84.2%        Very Good      YES
Samuel Chukwueze         Fulham  eng Premier League  26.0          18.0  84.1%        Very Good      YES
 Florian Thauvin           Lens          fr Ligue 1  32.0           5.0  84.0%        Very Good      YES
    Jarrod Bowen West Ham United  eng Premier League  28.0          30.0  82.1%        Very Good

WHY Harry Wilson is the closest match (91.3% - Excellent Match)
Most alike in these stats:
   Shots per 90 | Shots on Target per 90 | Shots on Target % |
   Plus-Minus per 90 | Fouls Committed per 90

Search player: quit  ->  Goodbye!
```

```
SHAP Player Explainer     Enter a player name to explain their predicted market value (type 'quit' or 'q' to exit).
Search player: amad diallo
Predicted Market Value: EUR 59.4M     Actual Market Value: EUR 45.0M     Verdict: UNDERVALUED by EUR 14.4M
TOP FACTORS AFFECTING THE PREDICTION
Feature                    Player value   Effect      SHAP Value
League_eng Premier League         1.00   Increases         0.44
Team_Goals_On_Pitch              52.00   Increases         0.38
Years_Left                        4.49   Increases         0.34
Age                              23.00   Increases         0.32
Plus_Minus                       19.00   Increases         0.20
Unused_Sub                        0.00   Increases         0.16
Points_Per_Match                  2.03   Increases         0.15
On_Off_Impact                     0.73   Decreases        -0.13
Search player: q  ->  Goodbye!
```